

Clinical and MRI Markers for Acute vs Chronic Temporomandibular Disorders Using a Machine Learning and Deep Neural Networks

Corresponding Author: Professor Yeon-Hee Lee

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The article aims to study the impact of different factors on the chronicity of symptoms in patients with temporomandibular disorders (TMD) with machine learning techniques.

This paper highlights the differences of used machine learning techniques, emphasizing that logistic regress as conventional machine, and multilayer perceptron as deep learning. In fact, this article mis-interpret the basic concepts of machine learning and deep learning. Multilayer perceptron is one of the traditional but fundamental classic fully connected neural networks while deep learning usually refers to a suite of machine learning techniques including convolutional neural networks, long short-term memory (LSTM), transformer etc. Due to the limitation of fully connected neural networks in learning local features and attention, deep learning and transformers are proposed. This paper actually compares both conventional machine learning techniques logistic regression and multilayer perceptron. It is known that multilayer perceptron usually performs worse than popular deep learning and even support vector machines (SVM) in many cases. None of popular deep learning techniques have been mentioned and compared in this article while the paper claims to use deep learning.

On the other hand, possibly the performance and more convincing conclusions could be made if solid and latest machine learning techniques could be used.

Reviewer #2

(Remarks to the Author)

The manuscript utilizes statistical methods, machine learning (ML), and deep learning (DL) models to analyze the clinical, behavioral, and MRI characteristics of acute and chronic temporomandibular disorders (TMD) and identifies key predictors of chronicity.

While the study demonstrates innovation in combining AI techniques to analyze factors contributing to TMD chronicity, there are several critical methodological and result presentation issues that require substantial revision. Below are the detailed comments:

1. Clarification on the 19 features:

The study mentions the use of 19 features but does not clearly define or specify their composition. It is essential to provide a complete list of these features and clarify whether they include the four MRI findings (ADD, TMJ-OA, joint space narrowing, effusion). Additionally, it is unclear if "symptom duration" (as shown in Table 1) is included as one of the clinical features. If it is included, this introduces significant bias because "symptom duration >6 months" is a diagnostic criterion for chronic TMD. This overlap may compromise the independence of the feature and the validity of the model. Please specify the features comprehensively and ensure they are appropriately selected.

2. Characteristics and use of STOP-Bang features:

The role of the STOP-Bang score in the analysis is insufficiently described. The following aspects need clarification:

Are the individual STOP-Bang features (e.g., snoring, observed apnea) used as independent variables, or is the total score treated as a single feature?

Alternatively, is a specific threshold (e.g., STOP-Bang score ≥ 5) used for classification?

A clear explanation of how STOP-Bang features are handled in the model is necessary to enhance the interpretability and reproducibility of the results.

3. Extraction of MRI features:

The study does not detail the method of extracting MRI features (ADD, TMJ-OA, joint space narrowing, effusion). If these features were annotated manually, please specify the criteria and the level of expertise of the annotators (e.g., radiologists or imaging specialists). Additionally, inter-rater reliability metrics (e.g., Kappa or ICC values) should be provided to ensure consistency and objectivity in feature extraction.

4. Definition of the MLP model as deep learning:

The manuscript defines a multi-layer perceptron (MLP) with four hidden layers as a deep learning model. This categorization is debatable. While theoretically, deep learning refers to models with multiple hidden layers, the complexity of deep learning typically involves more sophisticated architectures and larger datasets. Considering the limited dataset size and relatively simple network architecture, the justification for categorizing the MLP model as a deep learning model requires further elaboration. Alternatively, the term should be revised for accuracy.

5. Sample size and risk of overfitting:

The total sample size of 239 is relatively small for training, validating, and testing deep learning models. This raises concerns about potential overfitting. The manuscript does not provide sufficient details on how overfitting was addressed.

Specifically:

- 1) Was cross-validation (e.g., k-fold cross-validation) performed to evaluate the model's generalizability?
- 2) Were strategies such as regularization or data augmentation applied?
- 3) Learning curves for both training and validation loss should be presented to evaluate model performance and stability.
- 4) The authors should also discuss the limitations imposed by the small sample size and how these were mitigated during the analysis.

Reviewer #3

(Remarks to the Author)

Based on the manuscript you've provided, here's an example of a review decision as a reviewer:

Review Decision: Minor Revisions

Summary:

The manuscript titled "Clinical and MRI Markers for Acute vs Chronic TMD Using a Machine Learning and Deep Learning Approach" addresses an important issue in temporomandibular disorders (TMD) by identifying factors that contribute to the progression from acute to chronic TMD using machine learning (ML) and deep learning (DL). The study is robust in its use of clinical data, MRI findings, and AI-based models to predict chronic TMD. The use of AI models, including logistic regression and multi-layer perceptron (MLP), is particularly noteworthy for predicting chronicity based on clinical and MRI markers.

Strengths:

Novelty and Relevance: The study tackles an important clinical question with modern AI-based approaches, adding value to the field of TMD research.

Comprehensive Dataset: The study includes a large cohort of 239 patients, providing a solid foundation for the findings.

Clear Methodology: The methodology is clearly described, with appropriate use of statistical and machine learning techniques for analysis.

Practical Implications: The results can have important clinical implications for early detection and intervention in TMD, which could improve patient care.

Weaknesses and Areas for Improvement:

Depth of Discussion on AI Model Comparison:

While the manuscript presents a comparison between logistic regression and MLP, the difference in AUROC (3.99%) is not statistically significant ($p = 0.3067$). This should be discussed further to justify the inclusion of both models, especially if the difference in performance is marginal. A more in-depth discussion on why MLP was chosen, despite the lack of significant improvement over logistic regression, would help the reader understand the trade-offs.

Clarification of MRI Analysis:

The manuscript mentions that effusion was more frequent in chronic TMD but not statistically significant ($p = 0.169$). While effusion may not be a distinguishing predictor, it would be useful to elaborate on its role and potential confounding effects in MRI findings.

Sleep Time as a Predictor:

Sleep time showed predictive power with a cut-off value of 6.750 hours (AUC = 0.614). This result may require further validation. It would be valuable to compare this sleep-related cut-off with established clinical thresholds or to explore how

this metric aligns with existing research on TMD and sleep disturbances.

Addressing Potential Confounding Factors:

Psychological stress and microtrauma were not significantly different between acute and chronic TMD, yet these factors are often linked to chronic pain conditions. While the lack of statistical significance is noted, further discussion on why these factors might not be predictive in this context could strengthen the manuscript.

Minor Revisions Needed:

Expand the discussion on the practical significance of the AI models (logistic regression vs. MLP) and justify their use despite minimal performance differences.

Provide more insight into the role of effusion in the progression of TMD and how it might be integrated into the prediction model.

Clarify the clinical relevance of the sleep time cut-off and compare it with existing literature on sleep-related disorders and TMD.

Consider a more in-depth exploration of potential confounding factors, especially regarding stress and microtrauma.

Recommendation: I recommend minor revisions before acceptance. The study is strong and contributes valuable insights, but addressing the points above will improve the clarity and significance of the findings.

Reviewer #4

(Remarks to the Author)

This manuscript investigates the differentiation between acute and chronic temporomandibular disorders (TMD) by utilizing clinical, sleep-related, and MRI findings, along with machine learning (ML) and deep learning (DL) models for prediction. The major claims include: Chronic TMD is significantly associated with bruxism, TMJ noise, sleep disturbances, and structural abnormalities such as anterior disc displacement (ADD), TMJ osteoarthritis (OA), and joint space narrowing.

The ML and DL models provide valuable predictive insights, with deep learning slightly improving upon logistic regression performance, though not significantly.

These claims are relevant and contribute to the field by combining clinical assessments with advanced AI techniques. The integration of ML/DL into TMD diagnostics is a novel and promising approach that could enhance predictive accuracy in clinical settings.

The manuscript presents a novel application of AI in differentiating acute from chronic TMD. While previous studies have explored MRI-based and clinical predictors of chronic TMD, the use of deep learning and machine learning in this context adds a valuable dimension. The findings are likely to be of interest to both clinicians and researchers in the fields of oral medicine, radiology, and computational healthcare.

Robust Methodology: The study employs a well-structured approach combining statistical analysis with AI techniques, providing comprehensive insights into TMD chronicity.

Clinical Relevance: The identification of key predictors can directly impact clinical decision-making and patient management.

Comparative Analysis: The comparison between logistic regression and deep learning adds depth to the study, showcasing the advantages and limitations of both approaches.

Visualization Techniques: The use of 2D and 3D visualization enhances the interpretability of complex interrelationships.

While the manuscript is well-structured and provides valuable insights, some areas require minor revisions:

Statistical Analysis Validation:

The manuscript does not provide a detailed justification for the choice of specific cut-off values in logistic regression (e.g., sleep time cut-off at 6.75 hours). Providing additional context or citing relevant literature would strengthen the argument.

The comparison between logistic regression and deep learning could benefit from additional statistical tests (e.g., McNemar's test) to confirm that the performance difference is not significant.

Reproducibility of the Study:

While the study provides GitHub access to code, further details on hyperparameter tuning and data preprocessing could improve reproducibility.

More specifics on MRI image acquisition settings (e.g., resolution, slice thickness) would help standardize replication in future studies.

Interpretability of Deep Learning Model:

While the logistic regression model provides clear feature weights, deep learning remains a "black box." Including techniques such as SHAP (SHapley Additive exPlanations) or feature importance visualization for the MLP model would improve interpretability.

The manuscript would benefit from citing relevant recent work, particularly DOI 10.1111/joor.13529, which explores related concepts in TMD chronicity and could strengthen the literature review.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Quality of the paper has been improved and all the comments have been addressed satisfactorily.

Reviewer #2

(Remarks to the Author)

The authors have addressed all of my concerns.

Reviewer #3

(Remarks to the Author)

The manuscript is strong, with clear relevance and contribution to clinical research in the context of TMD. It presents an innovative use of machine learning to identify chronicity predictors. However, further clarification on the model architecture, handling of overfitting, and the explanation of clinical variables (like the STOP-Bang score and MRI features) will improve the rigor of the study. The manuscript has been revised well in response to reviewer comments, and the overall research has potential clinical implications.

Recommendation: It can be accepted with minor revisions, primarily focusing on:

Expanding the discussion of model comparison: Providing a more detailed explanation of the small performance difference between logistic regression and MLP models, including why both models were included despite the marginal performance gap.

Refining explanations of data handling and model choices: Further clarifying the handling of the STOP-Bang score, the manual annotation process for MRI features, and the model architecture to avoid confusion and ensure transparency.

Reviewer #4

(Remarks to the Author)

Dear Authors,

Congratulations on your excellent and timely study, "Clinical and MRI Markers for Acute vs Chronic TMD Using a Machine Learning and Deep Neural Networks." It was a pleasure to read such a well-organized and thoughtful piece of work. The integration of clinical, behavioral, and MRI data with both conventional statistics and deep learning approaches is impressive and highly relevant to current advances in medical research. Your careful attention to model interpretability and clinical applicability is also much appreciated.

While the manuscript is very strong overall, I would like to suggest a few points that could help further refine and strengthen your work:

1. Methodology:

Data Handling: It would be beneficial to specify how any missing data were treated during preprocessing. Although this may not have been a major issue, a brief clarification would reassure readers about data completeness and model robustness.

External Validation: The study relies on internal cross-validation. Although you discuss model stability well, considering or acknowledging the value of an external validation set would enhance confidence in the generalizability of your findings.

Binary Classification of Chronicity: Using the 6-month cut-off is consistent with clinical norms, but perhaps a short discussion on the potential limitations of dichotomizing a continuous symptom duration variable could add nuance.

2. Statistical Analysis:

Given the number of variables analyzed, it might be worth discussing (even briefly) whether corrections for multiple comparisons beyond the Bonferroni adjustment for χ^2 tests were considered, particularly for continuous variables. This would help strengthen the robustness of the reported associations.

3. Tables:

In Table 1, it could be helpful to clearly label non-significant results (e.g., marking with "NS" or similar) to allow readers to quickly distinguish between significant and non-significant findings without needing to scan p-values individually.

Overall, I found this study highly valuable, and it advances our understanding of chronic TMD using a modern, multidisciplinary approach. I look forward to seeing the final version and future work building on these findings. Thank you for your important contribution to the field.

Version 2:

Reviewer comments:

Reviewer #3

(Remarks to the Author)

The authors have addressed all of comments.

Reviewer #4

(Remarks to the Author)

Dear Authors,

all the requested changes and suggestions have been carefully addressed and incorporated into the manuscript. After reviewing the updated version, I believe everything is now in place and the article is ready to move forward. Thanks again for your work on the revisions.

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Reviewers' comments:

I (Yone-Hee Lee), along with my co-authors, would like to sincerely thank the reviewers and the editor for their thoughtful and constructive feedback on our manuscript. We truly appreciate the time and effort you have devoted to evaluating our work, and we are grateful for the opportunity to revise and improve our submission. Your comments have been invaluable in helping us clarify key aspects of our study. We have made every effort to thoroughly address all the concerns and suggestions raised in the reviews. The authors have carefully revised the manuscript based on the reviewers' comments, and the revised sections are highlighted in red. We have addressed all the reviewers' comments and recommendations, and the online link to the code has also been updated accordingly (https://github.com/SeonggwangJeon/Chronic_VAS_visualization/tree/main). We kindly ask for your review. We would like to express our sincere gratitude to all of you.

Reviewer #1 (Remarks to the Author):

The article aims to study the impact of different factors on the chronicity of symptoms in patients with temporomandibular disorders (TMD) with machine learning techniques.

This paper highlights the differences of used machine learning techniques, emphasizing that logistic regress as conventional machine, and multilayer perceptron as deep learning. In fact, this article mis-interpret the basic concepts of machine learning and deep learning. Multilayer perceptron is one of the traditional but fundamental classic fully connected neural networks while deep learning usually refers to a suite of machine learning techniques including convolutional neural networks, long short-term memory (LSTM), transformer etc. Due to the limitation of fully connected neural networks in learning local features and attention, deep learning and transformers are proposed. This paper actually compares both conventional machine learning techniques logistic regression and multilayer perceptron. It is known that multilayer perceptron usually performs worse than popular deep learning and even support vector machines (SVM) in many cases. None of popular deep learning techniques have been mentioned and compared in this article while the paper claims to use deep learning.

On the other hand, possibly the performance and more convincing conclusions could be made if solid and latest machine learning techniques could be used.

Response:

We thank the reviewer for this insightful and constructive comment regarding our use of machine learning terminology and the architecture of our neural network model. We fully acknowledge the concern that referring to a basic multilayer perceptron (MLP) as “deep learning” may have been conceptually imprecise in our initial submission.

In response, we have revised the manuscript to avoid labeling the original MLP model as “deep learning” and instead describe it more accurately as a **simple feedforward neural network**. To address the reviewer’s suggestion for a more robust model, we conducted new experiments using an upgraded **five-layer deep neural network (DNN)** architecture, including **dropout**

layers for regularization, to better reflect deep learning standards. These modifications aimed to enhance the model's expressive power and reduce overfitting.

Moreover, we conducted **5-fold cross-validation**, applied **L2 regularization** for logistic regression, and included **learning curves** for both models to visualize training and validation loss. Despite these efforts, the performance gap between the DNN and logistic regression remained modest, suggesting no statistically significant improvement.

We have also incorporated **SHAP value analyses** to enhance the interpretability of the DNN model and provide clear explanations of feature contributions. Further, we have added a **comprehensive section detailing hyperparameter tuning**, data preprocessing strategies, and the rationale behind final model selection based on validation loss.

While we agree that models such as CNNs, LSTMs, or transformers represent more advanced deep learning architectures, we considered their application less suitable in our current study due to the structure of our tabular clinical and MRI-derived data. Nonetheless, we have now clarified this point in the revised discussion and acknowledged it as a limitation of the current approach. We also included a suggestion for applying more complex deep learning architectures in future work, especially for direct MRI image analysis.

As per your recommendation, the title of our study has been revised to: "Clinical and MRI Markers for Acute vs Chronic TMD Using Machine Learning and Deep Neural Networks." This revised title more accurately reflects the methodologies employed and avoids potential misinterpretation regarding the complexity of the models used.

We greatly appreciate your feedback, which allowed us to improve the rigor, accuracy, and clarity of our study.

Reviewer #2 (Remarks to the Author):

The manuscript utilizes statistical methods, machine learning (ML), and deep learning (DL) models to analyze the clinical, behavioral, and MRI characteristics of acute and chronic temporomandibular disorders (TMD) and identifies key predictors of chronicity.

While the study demonstrates innovation in combining AI techniques to analyze factors contributing to TMD chronicity, there are several critical methodological and result presentation issues that require substantial revision. Below are the detailed comments:

1. Clarification on the 19 features:

The study mentions the use of 19 features but does not clearly define or specify their composition. It is essential to provide a complete list of these features and clarify whether they include the four MRI findings (ADD, TMJ-OA, joint space narrowing, effusion). Additionally, it is unclear if "symptom duration" (as shown in Table 1) is included as one of the clinical features. If it is included, this introduces significant bias because "symptom duration >6 months" is a diagnostic criterion for chronic TMD. This overlap may compromise the independence of the feature and the validity of the model. Please specify the features comprehensively and ensure they are appropriately selected.

Response:

We sincerely thank the reviewer for this thoughtful and important comment. In response, we have clarified the composition of the 19 features in the revised Statistical Analysis section of the manuscript. Specifically, the 19 clinical characteristics and MRI findings used to predict chronic TMD, as opposed to acute TMD, included 15 clinical variables—sex, age, visual analog scale (VAS) score, TMJ noise, TMD-related pain, sleep time, joint locking, muscle stiffness, bruxism, macrotrauma, tinnitus, psychological distress, sleep problems, and STOP-Bang score thresholds (≥ 3 and ≥ 5)—and 4 MRI findings: anterior disc displacement (ADD), temporomandibular joint osteoarthritis (TMJ-OA), effusion, and joint space narrowing.

Contrary to your concern, we confirm that "symptom duration" was not included as a feature in the model. We fully agree that including symptom duration as a predictor would have introduced circular reasoning, as it is part of the diagnostic criteria for chronic TMD. We greatly appreciate your careful review, which helped us to ensure the transparency and rigor of our methodology.

2. Characteristics and use of STOP-Bang features:

The role of the STOP-Bang score in the analysis is insufficiently described. The following aspects need clarification:

Response:

We appreciate the reviewer's question. In our predictive modeling, we did **not use the individual STOP-Bang components** (e.g., snoring, observed apnea, BMI, age, etc.) as separate input features. Although these variables were collected and presented in **Table 2** for descriptive purposes, they were excluded from the final model to avoid unnecessary increase in feature dimensionality, which could lead to overfitting given our sample size.

Are the individual STOP-Bang features (e.g., snoring, observed apnea) used as independent variables, or is the total score treated as a single feature?

Response:

Yes, we used the **total STOP-Bang score** as the basis for feature construction in the model. This approach is supported by previous literature, where the total score has been shown to be a reliable indicator of sleep apnea risk. We did not use the raw total score as a continuous variable, but instead applied clinically validated thresholds to create binary indicators, as explained below.

Alternatively, is a specific threshold (e.g., STOP-Bang score ≥ 5) used for classification?

A clear explanation of how STOP-Bang features are handled in the model is necessary to enhance the interpretability and reproducibility of the results.

Response:

Indeed, we used two **clinically established cut-off points** for the STOP-Bang score to create **two binary variables**:

- **STOP-Bang score ≥ 3** to indicate intermediate risk of OSA
- **STOP-Bang score ≥ 5** to indicate high risk of OSA

This decision was based on established clinical guidelines and supported by previous studies (e.g., Chung et al., 2008; Nagappa et al., 2017). By including both thresholds, we aimed to capture a broader range of OSA risk that may be associated with chronic TMD development.

The STOP-Bang questionnaire consists of eight dichotomous (yes/no) items that assess clinical features associated with obstructive sleep apnea (OSA). Each affirmative (“yes”) response scores 1 point, while a negative (“no”) response scores 0, resulting in a total score ranging from 0 to 8. Based on established clinical guidelines, a total score of ≥ 3 indicates an intermediate risk of OSA, while a score of ≥ 5 indicates a high risk.

In our study, while the individual STOP-Bang items and their distributions are presented in Table 2 for descriptive purposes, only the total score was used in the predictive modeling of chronic TMD. Specifically, two binary variables were derived from the total score: one indicating a score ≥ 3 (intermediate risk) and the other indicating a score ≥ 5 (high risk). This approach was chosen to reduce the dimensionality of the feature space and to mitigate the risk of model overfitting, given the limited sample size. All participants completed the STOP-Bang questionnaire, and self-reported average sleep time (in hours) over the past two weeks was also collected and included as a separate variable.

3. Extraction of MRI features:

The study does not detail the method of extracting MRI features (ADD, TMJ-OA, joint space narrowing, effusion). If these features were annotated manually, please specify the criteria and the level of expertise of the annotators (e.g., radiologists or imaging specialists). Additionally, inter-rater reliability metrics (e.g., Kappa or ICC values) should be provided to ensure

consistency and objectivity in feature extraction.

Response:

We appreciate the reviewer's important comment regarding the MRI feature extraction process. The imaging features—anterior disc displacement (ADD), temporomandibular joint osteoarthritis (TMJ-OA), effusion, and joint space narrowing—were manually annotated based on clinical MRI interpretations.

The initial reference labels were determined through a consensus by radiology specialists at Kyung Hee University Hospital, who conducted multiple rounds of image review to ensure diagnostic accuracy. Subsequently, two co-authors of this study, Dr. Yeon-Hee Lee and Dr. Q-Schick Auh—both certified specialists in orofacial pain and oral medicine in South Korea with over 8 years of experience in MRI interpretation and TMD diagnosis—independently reviewed the MRIs. This approach has also been adopted in our previous studies.

To assess inter-rater reliability, the annotations provided by the two authors were compared with the initial radiologist consensus. Since the assessments involved binary decisions, we have revised the expressions of intra-examiner and inter-examiner reproducibility in the manuscript to reflect Cohen's Kappa coefficients instead of intra-class correlation coefficients (ICCs). Intra-examiner reproducibility yielded Cohen's kappa values of 0.78 and 0.85, while inter-examiner kappa values for TMJ effusion diagnosis were 0.79 and 0.83, respectively. Any disagreements were resolved through discussion until a consensus was reached. Any discrepancies were resolved through consensus after multiple rounds of discussion, and the final agreed-upon labels were used in the analysis.

We have now included this detailed explanation in the revised Methods section to enhance the transparency and reproducibility of our imaging feature extraction process.

We sincerely appreciate your thoughtful feedback. At the same time, we kindly ask for your understanding that conducting real-world research often involves practical limitations and challenges. Nonetheless, we are truly grateful for your insights, which have helped us improve the clarity and rigor of our study.

4. Definition of the MLP model as deep learning:

The manuscript defines a multi-layer perceptron (MLP) with four hidden layers as a deep learning model. This categorization is debatable. While theoretically, deep learning refers to models with multiple hidden layers, the complexity of deep learning typically involves more sophisticated architectures and larger datasets. Considering the limited dataset size and relatively simple network architecture, the justification for categorizing the MLP model as a deep learning model requires further elaboration. Alternatively, the term should be revised for accuracy.

Response:

Thank you for your thoughtful observation. We agree that the term “deep learning” typically refers to more complex architectures such as convolutional neural networks (CNNs), recurrent

neural networks (RNNs), or transformers, often trained on large-scale datasets. In the original version of the manuscript, we employed a three-layer MLP, which may not have met the conventional criteria for deep learning.

In the revised manuscript, we have updated our terminology and description for clarity and accuracy. We now refer to our enhanced model as a **deep neural network (DNN)** rather than simply “deep learning,” and we have implemented a five-layer architecture with dropout regularization to better reflect the depth and non-linearity typically associated with deep learning models.

Furthermore, we explicitly acknowledge the simplicity of the model in comparison to more advanced architectures and have included a discussion of this limitation in the revised Discussion section. This distinction helps avoid conceptual misrepresentation while maintaining the integrity of our comparative analysis.

5. Sample size and risk of overfitting:

The total sample size of 239 is relatively small for training, validating, and testing **deep learning** models. This raises concerns about potential overfitting. The manuscript does not provide sufficient details on how overfitting was addressed.

Response:

We appreciate the reviewer’s important concern regarding the potential risk of overfitting due to the limited sample size. In response, we have taken several steps to mitigate overfitting and improve model generalizability:

1. **Dropout Regularization in DNN:** Dropout layers were incorporated into the DNN architecture to prevent overfitting by randomly deactivating neurons during training.
2. **L2 Regularization in Logistic Regression:** To ensure fairness and improve generalizability, we also applied L2 regularization (weight decay) in the logistic regression model.
3. **Cross-Validation Strategy:** We implemented **5-fold cross-validation** for both models. This reduced bias from a single train-test split and allowed for a more robust estimation of model performance. Final AUROC values were derived from the **fold-wise averaged outputs**.
4. **Train-Validation-Test Split:** We revised our data splitting strategy from the original 80:20 to a **60:20:20 split** (train:validation:test), enabling better parameter tuning without overfitting to the test set.
5. **Learning Curves Added:** Training and validation loss curves were included to visually evaluate the model’s learning behavior, showing no signs of severe overfitting.
6. **Early Stopping:** Model parameters were selected based on the epoch with the **lowest validation loss**, ensuring generalizability.

We have described these strategies in the revised **Methods section** and discussed the inherent limitation of sample size explicitly in the **Discussion section**.

Specifically:

1) Was cross-validation (e.g., k-fold cross-validation) performed to evaluate the model's generalizability?

Response:

Yes, we performed **5-fold cross-validation** for both the logistic regression and deep neural network (DNN) models. This approach allowed us to obtain more robust and unbiased performance estimates by reducing the dependency on a single train-test split. The final AUROC values were calculated by averaging the fold-wise outputs.

2) Were strategies such as regularization or data augmentation applied?

Response:

To mitigate overfitting, we applied **dropout regularization** within the DNN architecture and **L2 regularization (weight decay)** in the logistic regression model. These strategies help enhance generalizability by preventing the models from relying too heavily on specific patterns in the training data. We did not employ data augmentation, as our input data were structured (tabular) and not image-based.

3) Learning curves for both training and validation loss should be presented to evaluate model performance and stability.

Response:

We appreciate your valuable feedback. Figure 7 has been added, and a corresponding explanation has been incorporated into the Results section. Thank you.

To assess training stability and generalizability, loss curves from 5-fold cross-validation were analyzed (Figure 7). For logistic regression (Fig. 7a–7e), training loss consistently decreased, while validation loss declined initially and then slightly increased—a typical learning pattern. The dashed lines mark the epochs with minimum validation loss, which were used for model selection. The consistency across folds indicates stable training without underfitting or overfitting.

For the deep neural network (Fig. 7f–7j), training loss steadily decreased, and validation loss showed an early drop followed by a mild rise, suggesting slight overfitting. Early stopping based on minimum validation loss helped prevent severe overfitting. These results support the robustness of both models.

4) The authors should also discuss the limitations imposed by the small sample size and how these were mitigated during the analysis.

Response:

While our sample size of 239 patients exceeded the minimum required based on power analysis, we acknowledge that it remains relatively modest for deep learning applications. To address this, we implemented several strategies including **regularization, a 60:20:20 train-**

validation-test split, and **cross-validation** as mentioned above. Additionally, model selection was based on the epoch with the **lowest validation loss** (early stopping) to avoid overfitting. These steps helped us ensure the reliability of our findings despite the limited dataset size.

We have revised the **Methods**, **Results**, and **Discussion** sections accordingly to clearly describe these methodological refinements and limitations.

Reviewer #3 (Remarks to the Author):

Based on the manuscript you've provided, here's an example of a review decision as a reviewer:

Review Decision: Minor Revisions

Response:

We deeply appreciate your recognition of the value and contribution of our research.

Summary:

The manuscript titled "Clinical and MRI Markers for Acute vs Chronic TMD Using a Machine Learning and Deep Learning Approach" addresses an important issue in temporomandibular disorders (TMD) by identifying factors that contribute to the progression from acute to chronic TMD using machine learning (ML) and deep learning (DL). The study is robust in its use of clinical data, MRI findings, and AI-based models to predict chronic TMD. The use of AI models, including logistic regression and multi-layer perceptron (MLP), is particularly noteworthy for predicting chronicity based on clinical and MRI markers.

Strengths:

Novelty and Relevance: The study tackles an important clinical question with modern AI-based approaches, adding value to the field of TMD research.

Comprehensive Dataset: The study includes a large cohort of 239 patients, providing a solid foundation for the findings.

Clear Methodology: The methodology is clearly described, with appropriate use of statistical and machine learning techniques for analysis.

Practical Implications: The results can have important clinical implications for early detection and intervention in TMD, which could improve patient care.

Weaknesses and Areas for Improvement:

Depth of Discussion on AI Model Comparison:

While the manuscript presents a comparison between logistic regression and MLP, the difference in AUROC (3.99%) is not statistically significant ($p = 0.3067$). This should be discussed further to justify the inclusion of both models, especially if the difference in performance is marginal. A more in-depth discussion on why MLP was chosen, despite the lack of significant improvement over logistic regression, would help the reader understand the trade-offs.

Response:

We appreciate this important observation. In the revised manuscript, we now provide additional discussion on the comparative utility of logistic regression and deep neural networks. While the AUROC improvement of the DNN was not statistically significant, we believe including both models offers practical insights. Logistic regression serves as a well-understood baseline that is easily interpretable in clinical settings, whereas the DNN provides an opportunity to capture nonlinear relationships among features. In many clinical contexts, even modest performance gains may be meaningful when balanced with model complexity and interpretability. This trade-off is now explicitly discussed in the revised Discussion section:

From a diagnostic perspective, both the logistic regression and DNN models exhibited high performance in predicting chronic TMD, demonstrating their potential utility in clinical practice. Although the DNN achieved a slightly higher AUROC compared to logistic regression, the difference was not statistically significant (DeLong's test). Despite this modest performance gap, the inclusion of both models provides practical insights into their respective advantages. Their strength lies in their ease of interpretation, which is crucial for clinicians making both time-sensitive and time-consuming decisions²⁵. These tools can enhance patient outcomes by minimizing diagnostic errors and improving decision-making efficiency, thus enhancing patient outcomes. Logistic regression is highly interpretable, easily implemented, and well-aligned with clinical decision-making workflows²⁶. It produces transparent coefficients that clinicians can readily interpret and use to guide patient care.

In contrast, the DNN captures non-linear relationships and complex interactions among features that may be overlooked by conventional statistical models²⁷. The use of both models in this study reflects a deliberate trade-off between interpretability and predictive power—an issue increasingly emphasized in medical AI research. While DNNs may provide incremental gains in performance, particularly when larger and more complex datasets are available, the transparency and clinical familiarity of logistic regression remain valuable in real-world healthcare settings^{28,29}. As deep learning methods continue to evolve and gain acceptance in clinical applications, traditional models like logistic regression still offer an optimal balance between performance and usability³⁰. Additionally, this study incorporated advanced 2D and 3D visualizations along with AI-based analysis, which further enhance the understanding of the multifactorial mechanisms contributing to chronic TMD.

Clarification of MRI Analysis:

The manuscript mentions that effusion was more frequent in chronic TMD but not statistically significant ($p = 0.169$). While effusion may not be a distinguishing predictor, it would be useful to elaborate on its role and potential confounding effects in MRI findings.

Response:

We agree with the reviewer that this point merits further clarification. Effusion, though not statistically significant in distinguishing between acute and chronic TMD ($p = 0.169$), may still reflect an underlying inflammatory response. However, our analysis found weak correlations between effusion and other structural MRI abnormalities, suggesting that effusion may be a nonspecific marker of chronic TMJ pathology. We have added this interpretation to the revised Discussion section.

This interpretation has been added to the revised Discussion section to address the reviewer's comment regarding the role of effusion:

In contrast, effusion, defined as fluid accumulation within the TMJ, was observed in a high proportion of patients (62.4% in acute TMD and 71.3% in chronic TMD). However, effusion did not emerge as a statistically significant predictor of chronicity in our model ($p = 0.169$). Despite its frequency, effusion demonstrated weak associations with other structural MRI abnormalities, such as ADD and joint space narrowing. This suggests that effusion may serve as a nonspecific marker of TMJ inflammation rather than a reliable indicator of structural

progression. Prior studies have also reported that effusion can be present in both symptomatic and asymptomatic individuals, limiting its diagnostic specificity^{25,26}. While effusion may reflect an active inflammatory response associated with TMD symptoms²⁷, its inconsistent correlation with joint degeneration indicates that it might act more as a transient phenomenon than a predictor of chronic TMD. These findings highlight the need for a more nuanced interpretation of effusion in MRI analyses and support the inclusion of multiple structural and clinical variables when assessing chronicity.

Sleep Time as a Predictor:

Sleep time showed predictive power with a cut-off value of 6.750 hours (AUC = 0.614). This result may require further validation. It would be valuable to compare this sleep-related cut-off with established clinical thresholds or to explore how this metric aligns with existing research on TMD and sleep disturbances.

Response:

Thank you for pointing this out. In the revised Discussion, we now compare the 6.75-hour threshold to findings in existing literature, which often suggest that <7 hours of sleep is associated with increased pain sensitivity and musculoskeletal complaints. Our findings align with this body of work, further supporting the role of reduced sleep duration as a contributor to chronic pain conditions, including TMD.

Epidemiological data from the National Health and Nutrition Examination have demonstrated that short sleep duration (<7 hours) is linked to a heightened risk of chronic musculoskeletal pain. Nonetheless, empirical research directly evaluating sleep duration among individuals with TMD is currently scarce.

Addressing Potential Confounding Factors:

Psychological stress and microtrauma were not significantly different between acute and chronic TMD, yet these factors are often linked to chronic pain conditions. While the lack of statistical significance is noted, further discussion on why these factors might not be predictive in this context could strengthen the manuscript.

Response:

We acknowledge the reviewer's point. Although psychological stress and microtrauma have been widely associated with chronic pain conditions, these variables did not reach statistical significance in our dataset. We suspect this may be due to self-report bias or limitations in measurement sensitivity, as both factors were based on binary survey items. We have added this explanation to the revised Discussion section.

Minor Revisions Needed:

Expand the discussion on the practical significance of the AI models (logistic regression vs. MLP) and justify their use despite minimal performance differences.

Provide more insight into the role of effusion in the progression of TMD and how it might be integrated into the prediction model.

Clarify the clinical relevance of the sleep time cut-off and compare it with existing literature on sleep-related disorders and TMD.

Consider a more in-depth exploration of potential confounding factors, especially regarding stress and microtrauma.

Recommendation: I recommend minor revisions before acceptance. The study is strong and

contributes valuable insights, but addressing the points above will improve the clarity and significance of the findings.

Response:

Your thoughtful comments and suggestions have been immensely constructive and have provided us with a hopeful and meaningful direction for our research. We sincerely affirm that we have made our best effort to revise and update the study in full consideration of all your feedback. Thank you once again for your valuable input.

Reviewer #4 (Remarks to the Author):

This manuscript investigates the differentiation between acute and chronic temporomandibular disorders (TMD) by utilizing clinical, sleep-related, and MRI findings, along with machine learning (ML) and deep learning (DL) models for prediction. The major claims include: Chronic TMD is significantly associated with bruxism, TMJ noise, sleep disturbances, and structural abnormalities such as anterior disc displacement (ADD), TMJ osteoarthritis (OA), and joint space narrowing.

The ML and DL models provide valuable predictive insights, with deep learning slightly improving upon logistic regression performance, though not significantly.

These claims are relevant and contribute to the field by combining clinical assessments with advanced AI techniques. The integration of ML/DL into TMD diagnostics is a novel and promising approach that could enhance predictive accuracy in clinical settings.

The manuscript presents a novel application of AI in differentiating acute from chronic TMD. While previous studies have explored MRI-based and clinical predictors of chronic TMD, the use of deep learning and machine learning in this context adds a valuable dimension. The findings are likely to be of interest to both clinicians and researchers in the fields of oral medicine, radiology, and computational healthcare.

Response:

We sincerely appreciate your constructive and academically insightful comments. Your thoughtful feedback has greatly contributed to improving the quality and clarity of our work. In order to ensure that your time and efforts were fully respected, we have made every effort to revise and update our manuscript accordingly. All modifications have been highlighted in red for your convenience. Thank you once again for your valuable input.

Robust Methodology: The study employs a well-structured approach combining statistical analysis with AI techniques, providing comprehensive insights into TMD chronicity.

Response:

To ensure methodological rigor, we employed a structured modeling pipeline combining conventional statistical analysis with artificial intelligence techniques. Logistic regression was used as a baseline due to its interpretability, while a five-layer DNN was implemented to capture potential non-linear associations among variables. Feature selection was guided by clinical relevance and variance inflation factor (VIF) analysis to minimize multicollinearity. Missing values were addressed using complete-case analysis, and model performance was assessed using AUROC, precision, recall, and F1-score. To further enhance interpretability, SHAP values were computed for the DNN. All modeling procedures were conducted in Python using Scikit-learn and PyTorch frameworks, and the complete codebase has been made available on GitHub to ensure reproducibility. Thank you.

Clinical Relevance: The identification of key predictors can directly impact clinical decision-

making and patient management.

Response:

We sincerely appreciate the reviewer's recognition of the clinical value of our study. Identifying key predictors such as bruxism, TMJ noise, sleep disturbances, and structural MRI findings (e.g., ADD, TMJ-OA) enables earlier recognition of patients at risk for chronic TMD. By highlighting these clinically actionable factors, our findings may guide personalized treatment strategies, promote early intervention, and improve long-term outcomes in TMD management. We have further emphasized these implications in the revised **Discussion** section:

This study demonstrated that the progression from acute to chronic TMD is influenced by a combination of clinical, behavioral, and structural factors. Many of these predictors—such as bruxism, TMJ noise, sleep disturbances, and MRI abnormalities—are either modifiable or clinically observable, making them practical markers for early identification of patients at risk for chronic progression. Timely recognition of these risk factors can guide personalized and targeted therapeutic interventions, thereby improving clinical outcomes. AI-based models, particularly DNNs, enhance the explanatory power of these predictors and support risk stratification. Although the DNN showed a slight improvement in predictive performance over logistic regression, traditional machine learning models remain valuable due to their simplicity and interpretability. These findings may help clinicians implement early intervention strategies to prevent symptom chronicity and enhance quality of life in patients with TMD.

Thank you.

Comparative Analysis: The comparison between logistic regression and deep learning adds depth to the study, showcasing the advantages and limitations of both approaches.

Response:

We sincerely appreciate the reviewer's thoughtful recognition of our comparative approach. The inclusion of both logistic regression and deep neural networks (DNNs) was intentional, allowing us to examine the trade-offs between model interpretability and predictive capacity. While DNNs can capture complex, nonlinear relationships and achieved slightly higher performance, logistic regression offers transparency and ease of interpretation—qualities essential in real-world clinical decision-making. We believe that presenting both models adds methodological depth and enhances the translational relevance of our findings.

[Introduction] -----

In this study, we aimed to identify clinical, sleep-related, and MRI-derived predictors of chronic TMD by comparing classical statistical methods with AI-based models. We employed logistic regression as a representative statistical and ML approach, and a deep neural networks (DNNs) with multiple hidden layers and regularization techniques as a more expressive non-linear model. While our DNN is relatively simple compared to advanced deep neural architectures such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), or transformer-based models, we designed it to be suitable for analyzing our structured tabular data¹³. This comparative analysis between logistic regression and DNN highlights the trade-off between interpretability and predictive power. The DNN demonstrated an ability to capture more complex feature interactions, whereas logistic regression offered greater

transparency and ease of clinical interpretation, which may facilitate broader clinical adoption. This model selection reflects a balance between feasibility, data structure, and clinical applicability. Through this approach, we aimed to improve the early detection of chronicity-related risk factors and provide clinical insights to support personalized treatment planning.

Visualization Techniques: The use of 2D and 3D visualization enhances the interpretability of complex interrelationships.

Response:

We appreciate the reviewer's recognition of our visualization efforts. To further enhance the interpretability of complex interrelationships, we improved the clarity and readability of the 2D and 3D figures in the revised manuscript. Additionally, we have included an online link to the interactive visualizations to facilitate easier exploration of the results. Thank you.

While the manuscript is well-structured and provides valuable insights, some areas require minor revisions:

Statistical Analysis Validation:

The manuscript does not provide a detailed justification for the choice of specific cut-off values in logistic regression (e.g., sleep time cut-off at 6.75 hours). Providing additional context or citing relevant literature would strengthen the argument.

Response:

We sincerely appreciate the reviewer's constructive and encouraging comment. In response, we have added a detailed explanation regarding the rationale for selecting the 6.75-hour sleep time cut-off in the revised manuscript. Additionally, we cited relevant literature supporting the association between short sleep duration and chronic musculoskeletal pain to strengthen the clinical relevance of this threshold. Thank you.

The comparison between logistic regression and deep learning could benefit from additional statistical tests (e.g., McNemar's test) to confirm that the performance difference is not significant (The McNemar test comparing the two models yielded a chi-square value of $\chi^2 = 1.125$ with a p-value of 0.289, indicating no statistically significant difference.). The corresponding content has been added to the Results section and incorporated into Figure 6.

Reproducibility of the Study:

While the study provides GitHub access to code, further details on hyperparameter tuning and data preprocessing could improve reproducibility.

Response:

Thank you for your helpful suggestion regarding reproducibility. To address this, we have added detailed descriptions of both hyperparameter tuning procedures and data preprocessing

steps in the revised Methods section. Specifically, we now describe the grid search ranges used for logistic regression and deep neural network (DNN) models, including learning rates, batch sizes, number of epochs, and regularization parameters. In addition, we clarified our data splitting strategy (60:20:20 for training, validation, and testing), normalization procedures, and the criteria used for selecting the best model based on validation loss. These updates, together with the GitHub repository containing our code and sample data structure, are intended to ensure transparency and enhance reproducibility for future research.

More specifics on MRI image acquisition settings (e.g., resolution, slice thickness) would help standardize replication in future studies.

Response:

Thank you for your thoughtful suggestion. We agree that clearly reporting MRI acquisition parameters is important for reproducibility and standardization in future research. In fact, our MRI protocol was well-established and consistently applied throughout the study. High-resolution MRIs were obtained using a 3T MRI scanner (Signa™ Genesis, GE Healthcare, Chicago, IL, USA) equipped with a 6 cm × 8 cm diameter surface coil. The scans included sagittal oblique sections (section thickness ≤3 mm, field of view 15 cm, matrix 256 × 224), with spin-echo sagittal images obtained on axial localizers. T2-weighted images were acquired using 650/14 and 2650/82 TR/TE sequences, while proton density images were acquired using a 2650/82 TR/TE protocol. These MRI protocols were performed in accordance with established standards at Kyung Hee University Medical Center and have been described in detail in prior publications [Refs. 2, 52]. We have now clarified this in the revised Methods section to improve transparency and facilitate future replication.

Interpretability of Deep Learning Model:

While the logistic regression model provides clear feature weights, deep learning remains a "black box." Including techniques such as SHAP (SHapley Additive exPlanations) or feature importance visualization for the MLP model would improve interpretability.

Response:

Thank you for this insightful comment. To improve the interpretability of our deep neural network (DNN) model, we incorporated SHAP (SHapley Additive exPlanations) analyses and generated feature importance visualizations. These visualizations present each feature's contribution to model predictions, summarized as fold-wise means ± standard deviations across the cross-validation folds. This addition allows for a more transparent understanding of the DNN's decision-making process and addresses the concern regarding its "black-box" nature.

The manuscript would benefit from citing relevant recent work, particularly DOI 10.1111/joor.13529, which explores related concepts in TMD chronicity and could strengthen the literature review.

Response:

We have added the following sentence to the revised manuscript and cited the reference you kindly suggested for your consideration:

“In a recent MRI study, chronic TMJ inflammation has been associated with disc anomalies and bony erosive or deformative changes.”

Reviewers' comments:

Reviewer #1 (Remarks to the Author):

Quality of the paper has been improved and all the comments have been addressed satisfactorily.

Response:

We would like to express our sincere gratitude for your positive evaluation of our revised manuscript.

Your thoughtful and encouraging review has provided our research team with significant motivation, deepened our commitment to scientific inquiry, and strengthened our confidence in advancing this line of research. All authors deeply respect and appreciate the time and expertise you have devoted to reviewing our work. Thank you once again for your invaluable contribution.

Reviewer #2 (Remarks to the Author):

The authors have addressed all of my concerns.

Response:

We sincerely appreciate your kind and supportive assessment of our revised manuscript.

Your generous academic judgment has greatly encouraged our research team, reaffirmed our passion for scientific investigation, and reinforced our confidence in pursuing further advancements.

All authors respectfully extend our heartfelt thanks for the time, effort, and expertise you have contributed to the review process.

Thank you very much for your valuable support.

Reviewer #3 (Remarks to the Author):

The manuscript is strong, with clear relevance and contribution to clinical research in the context of TMD. It presents an innovative use of machine learning to identify chronicity predictors.

Response:

We sincerely appreciate your valuable comments and suggestions.

Below, we have provided point-by-point responses to each of your remarks.

Additionally, all further revisions made during this second round of review have been highlighted in green within the manuscript.

However, further clarification on the model architecture, handling of overfitting, and the explanation of clinical variables (like the STOP-Bang score and MRI features) will improve the rigor of the study.

The concerns regarding model architecture, handling of overfitting, and the explanation of clinical variables have been thoroughly addressed in the revised manuscript.

Specifically, in the **Methods (AI-based evaluation and visualization)** section, we have provided a detailed description of the model architecture, including the use of a multi-layer perceptron (MLP) structure, the number of neurons per hidden layer, the application of ReLU activation functions, a 30% dropout rate, the use of the Adam optimizer, softmax activation for classification, and the implementation of early stopping based on validation loss. Thus, the reviewer's request for clarification on model architecture has been fully met. Regarding **handling of overfitting**, we employed several strategies, including dropout regularization, L2 weight decay, 5-fold cross-validation, and early stopping. In addition, training and validation loss curves were analyzed to confirm that no severe overfitting occurred. Therefore, the manuscript sufficiently addresses concerns about overfitting prevention and model robustness. Collectively, these revisions ensure that the reviewer's comments have been fully and appropriately addressed.

The manuscript has been revised well in response to reviewer comments, and the overall research has potential clinical implications.

Response:

We deeply appreciate your positive and constructive feedback.

Recommendation: It can be accepted with minor revisions, primarily focusing on:

Expanding the discussion of model comparison: Providing a more detailed explanation of the small performance difference between logistic regression and MLP models, including why both models were included despite the marginal performance gap.

Response:

Thank you for your insightful comment. We have now provided a more detailed explanation in the manuscript regarding the small performance difference between the logistic regression and MLP models, as well as the rationale for including both. Specifically, we chose to include

both models because, despite the marginal performance gap, they offer complementary advantages. Logistic regression provides a highly interpretable baseline with established clinical relevance, while the MLP model demonstrates the potential for capturing more complex, nonlinear patterns that may not be evident with conventional statistical methods. Including both models allows us to highlight the balance between model interpretability and potential performance gains, emphasizing the clinical applicability and methodological transparency of our findings. We have incorporated this explanation into the revised Discussion section (highlighted in green):

From a diagnostic perspective, both the logistic regression and DNN models exhibited high performance in predicting chronic TMD, demonstrating their potential utility in clinical practice. Although the DNN achieved a slightly higher AUROC (0.7708) compared to logistic regression (0.7083), the difference was not statistically significant. Despite this modest performance gap, the inclusion of both models provides practical insights into their respective advantages. To better understand the relative strengths and limitations of different modeling approaches, we compared logistic regression and MLP models. Logistic regression provides a clinically interpretable and robust baseline, whereas the MLP model captures more intricate patterns in the data, illustrating the trade-off between clinical applicability and methodological innovation^{30,31}. These complementary perspectives help minimize diagnostic errors and improve decision-making efficiency, ultimately enhancing patient outcomes³². Their strength lies in their ease of interpretation, which is crucial for clinicians making both time-sensitive and complex clinical decisions. Logistic regression, in particular, is highly interpretable, easily implemented, and well-aligned with clinical workflows³³. In contrast, the DNN model excels in capturing complex nonlinear relationships and feature interactions that may be overlooked by traditional statistical methods³⁴. It produces transparent coefficients that clinicians can readily interpret and use to guide patient care.

Refining explanations of data handling and model choices: Further clarifying the handling of the STOP-Bang score, the manual annotation process for MRI features, and the model architecture to avoid confusion and ensure transparency.

Response:

The code for all model architectures is available at the following link, which is referenced within the manuscript. In particular, the code for the DNN algorithm developed in this study to predict chronic TMD can be accessed via GitHub at https://github.com/SeonggwangJeon/Chronic_VAS_visualization/tree/main.

In terms of clinical variable explanations, the STOP-Bang questionnaire and MRI features were comprehensively described in the Methods (STOP-Bang and sleep time and MRI abnormal findings sections, respectively). The STOP-Bang questionnaire was explained in detail, including the eight items, scoring methods, and binary classifications based on clinical cut-offs (≥ 3 and ≥ 5 points).

MRI features, such as anterior disc displacement (ADD), temporomandibular joint osteoarthritis (TMJ-OA), joint space narrowing, and effusion, were clearly defined based on established diagnostic criteria.

Reviewer #4 (Remarks to the Author):

Dear Authors,

Congratulations on your excellent and timely study, "Clinical and MRI Markers for Acute vs Chronic TMD Using a Machine Learning and Deep Neural Networks." It was a pleasure to read such a well-organized and thoughtful piece of work. The integration of clinical, behavioral, and MRI data with both conventional statistics and deep learning approaches is impressive and highly relevant to current advances in medical research. Your careful attention to model interpretability and clinical applicability is also much appreciated.

Response:

We sincerely appreciate your thoughtful and encouraging comments regarding our manuscript.

We are truly honored by your recognition of the organization, relevance, and methodological integration of clinical, behavioral, and MRI data using both conventional statistical and deep learning approaches. Your acknowledgment of our efforts to ensure model interpretability and clinical applicability is deeply appreciated. Your positive feedback has been a great source of encouragement for our research team, and we are committed to continuing to advance our work in this field.

Thank you very much for your valuable time, careful review, and constructive support.

While the manuscript is very strong overall, I would like to suggest a few points that could help further refine and strengthen your work:

1. Methodology:

Data Handling: It would be beneficial to specify how any missing data were treated during preprocessing. Although this may not have been a major issue, a brief clarification would reassure readers about data completeness and model robustness.

Response:

Thank you for your insightful suggestion. In response, we have added a clarification in the Methods section regarding data handling. Participants with missing values in key clinical, behavioral, and imaging variables were excluded from the analysis. Minor missing data in auxiliary variables, if encountered, were handled using listwise deletion during preprocessing.

Study population

The study population comprised 239 consecutive patients with TMD (161 women and 78 men; mean age 35.60 ± 17.93 years) who visited Kyung Hee University Dental Hospital between January 2020 and September 2024. All diagnoses were made by a specialist with more than 10 years of clinical experience following the diagnostic criteria for TMD Axis I⁵¹. Patients with TMD were identified, and all clinical reports and MRI images of the TMJs were retrospectively reviewed. The duration of TMD symptoms, as reported by the patients, was recorded in months, with 6 months used as the threshold to classify the patients into two groups: acute TMD (symptom duration <6 months) and chronic TMD (symptom duration ≥ 6 months)⁵². This study compared the clinical and MRI characteristics associated with chronic TMD to those observed

in acute TMD and examined factors contributing to the prolonged duration of symptoms. The exclusion criteria were history of severe injuries, such as unstable multiple trauma to the orofacial area and maxillary and mandibular fractures; systemic diseases potentially affecting the TMJ, such as rheumatic diseases, systemic osteoarthritis, pregnancy, psychological problems, psychiatric or neurological disorders; and cases in which the structure of the TMJ complex was not clearly distinguishable on MRI ⁵³. Participants with missing values in key clinical, behavioral, and imaging variables were excluded from the analysis. Minor missing data in auxiliary variables, if encountered, were handled using listwise deletion during preprocessing.

External Validation: The study relies on internal cross-validation. Although you discuss model stability well, considering or acknowledging the value of an external validation set would enhance confidence in the generalizability of your findings.

Response:

Thank you for this valuable suggestion. We agree that external validation would further strengthen the generalizability of our findings. As our current study was based on a single-institution dataset, only internal cross-validation was performed. We have now added a statement acknowledging this limitation in the Discussion section and emphasized the need for future studies incorporating external validation across multiple centers.

Although internal cross-validation was performed to assess model stability, the generalizability of the findings could be further enhanced through external validation using independent, multi-center datasets. Future research should aim to replicate and extend these results across larger and more diverse populations.

Binary Classification of Chronicity: Using the 6-month cut-off is consistent with clinical norms, but perhaps a short discussion on the potential limitations of dichotomizing a continuous symptom duration variable could add nuance.

Response:

Thank you for the thoughtful suggestion. In response, we have added a brief discussion of the potential limitations associated with dichotomizing a continuous symptom duration variable. While using the 6-month cut-off aligns with clinical standards, we acknowledge that this approach may result in some loss of information regarding the spectrum of symptom duration. This limitation has been noted in the revised Discussion section.

Although prior studies have consistently associated psychological distress and microtrauma with the development and persistence of chronic pain conditions ^{46,47}, these factors did not reach statistical significance in our analysis. Psychological stress or distress is widely recognized as a contributor to pain chronification through mechanisms such as central sensitization and dysregulation of the hypothalamic-pituitary-adrenal axis ⁴⁸. Similarly, microtrauma—defined as repetitive minor injuries to the temporomandibular joint or masticatory muscles—has been suggested to play a role in the onset and persistence of TMD ⁹. One possible explanation is that the assessment of these variables relied solely on binary self-report items ⁴⁹, which may lack the sensitivity needed to capture the complexity and variability of individual experiences. Furthermore, while the use of a 6-month cut-off for symptom duration is consistent with clinical practice ⁵⁰, dichotomizing a continuous variable may have resulted in some loss of information and may not fully reflect the underlying spectrum of

chronicity. These methodological limitations may have led to an underestimation of the true prevalence or clinical impact of these factors within our cohort. Moreover, subjective responses are inherently susceptible to recall bias and underreporting, particularly in the context of chronic or multifactorial conditions⁵¹. Future research should incorporate validated psychometric instruments and detailed clinical assessments to more accurately characterize the role of psychological and mechanical factors in the chronicity of TMD.

2. Statistical Analysis:

Given the number of variables analyzed, it might be worth discussing (even briefly) whether corrections for multiple comparisons beyond the Bonferroni adjustment for χ^2 tests were considered, particularly for continuous variables. This would help strengthen the robustness of the reported associations.

Response:

Thank you for your valuable suggestion. In response, we have clarified in the Statistical Analysis section that Bonferroni correction was applied to adjust for multiple comparisons involving categorical variables (χ^2 tests), and false discovery rate control was applied to continuous variables to minimize the risk of type I error. We believe that this clarification strengthens the transparency and robustness of our statistical analysis.

Student's t-tests were used to compare the mean values between the two TMD groups, with false discovery rate control applied to adjust for multiple comparisons among continuous variables.

3. Tables:

In Table 1, it could be helpful to clearly label non-significant results (e.g., marking with "NS" or similar) to allow readers to quickly distinguish between significant and non-significant findings without needing to scan p-values individually.

Response:

Thank you very much for your thoughtful suggestion. We fully agree that clearly distinguishing significant and non-significant results can enhance the readability of the table. However, to maintain full transparency and allow readers to interpret the precise statistical outcomes themselves, we have decided to retain the original numerical p-values rather than replacing non-significant results with "NS." We believe that presenting the actual p-values preserves the integrity and openness of the data. We sincerely appreciate your valuable feedback and consideration.

Overall, I found this study highly valuable, and it advances our understanding of chronic TMD using a modern, multidisciplinary approach. I look forward to seeing the final version and future work building on these findings. Thank you for your important contribution to the field.

Thank you very much for your time and valuable feedback. We believe that your advice has significantly contributed to enhancing the quality of our study. We sincerely hope that this work will serve as a valuable resource for readers and gain wide interest. Once again, we deeply

appreciate your thoughtful review and support.

May 19, 2025

Editor-in-Chief

Communications Medicine

Nature Portfolio

Dear Editor,

We are pleased to resubmit our revised manuscript entitled “*Clinical and MRI Markers for Acute vs Chronic Temporomandibular Disorders Using a Machine Learning and Deep Neural Networks*” for consideration for publication in *Communications Medicine*.

In this study, we investigated clinical, behavioral, and MRI-based predictors of chronic temporomandibular disorders (TMD) by comparing logistic regression and a deep neural network (DNN) model. We conducted new experiments incorporating a five-layer DNN with regularization, cross-validation, and SHAP-based interpretability. Our findings highlight key risk factors—including bruxism, TMJ noise, and sleep disturbances—and demonstrate the potential utility of AI-based approaches in chronic pain prediction.

The authors have made every effort to complete the final version of the manuscript to the highest standard. We believe this work will be of interest to readers of *Communications Medicine* as it integrates clinical data with interpretable AI methods to inform early identification and intervention strategies for chronic pain conditions.

This manuscript is original, has not been published, and is not under consideration elsewhere. All authors have read and approved the final version.

We thank you for considering our revised submission and look forward to your response.

Sincerely,

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